

Processing

Composing with Algorithms
<http://www.bjarni-gunnarsson.net>

Recordings

Music derived from *recorded sounds* and the handling of these sounds.

The discussion will focus on:

- * *Recording sounds*
- * *Storing recorded sound*
- * *Playing back recorded sounds*
- * *Treatments of recorded sounds*

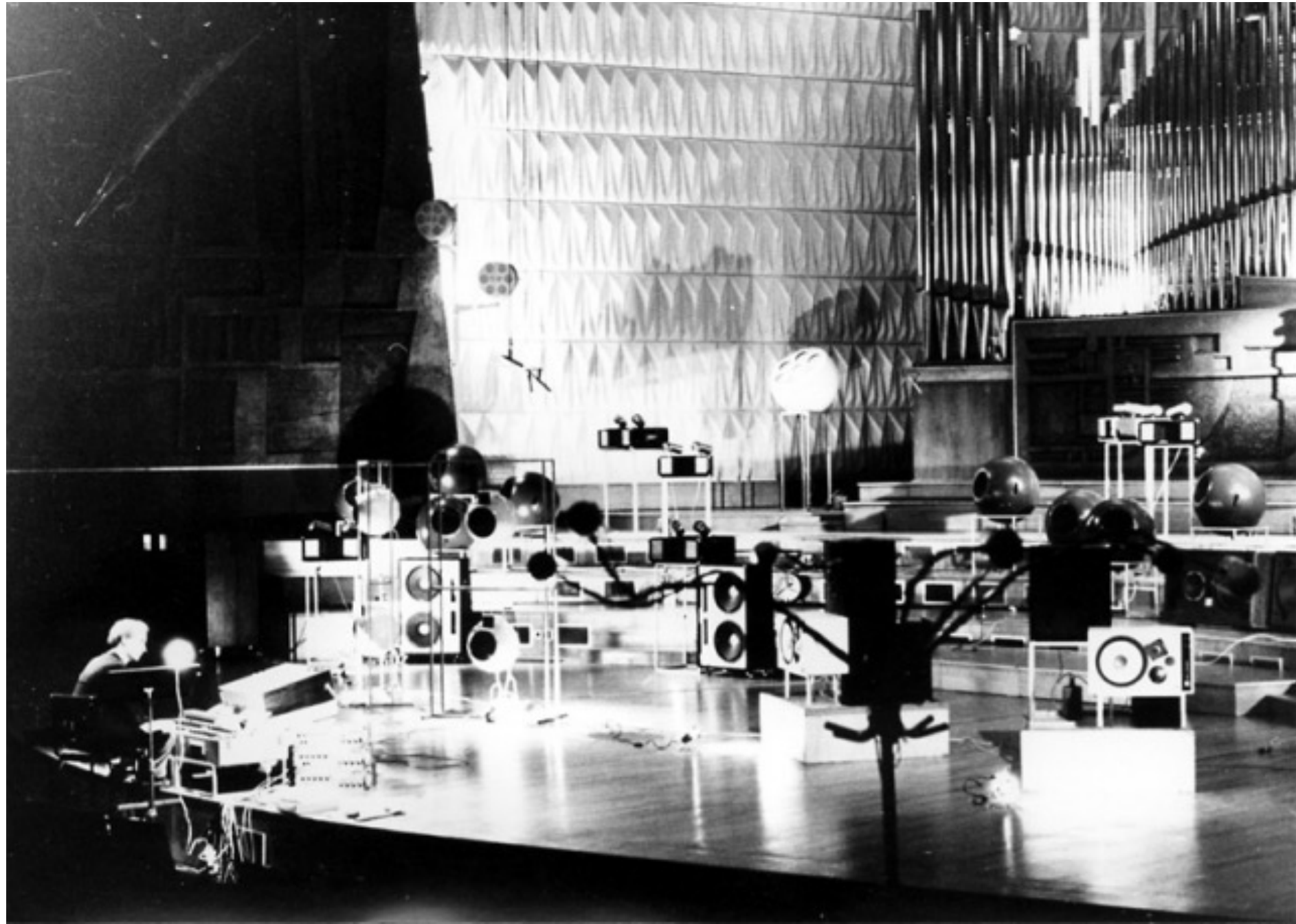
Musique concrete

Pierre Schaeffer started researching musical sound in 1942 with a focus on recording techniques and (1948) how to compose with recorded sounds.

Found that the ***timbre*** of sound is important but also its ***attack*** and ***decay*** characteristics.

Defined ***objet sonore*** as the pure sound event, isolated from its context and not possible to identify.

GRM Acousmonium



Acousmatic

“The term derives from Pythagoras, who gave lectures from behind a curtain so that his students would pay attention to the sound of his words. Today, the term “***acousmatic***” refers to compositions in which external reference or the hiding of it is central to the meaning of the work. A sound emanating from a loudspeaker emerges from behind a curtain, as it were, so we do not see the original source.”

“we encounter a distinction defined by Schaeffer as ***écoute réduite*** or reduced listening (Schaeffer 1966). Reduced listening means focusing on an event as a pure sound object, that is, in terms of its acoustic qualities, separate from its source and the meaning this may convey.”

(Curtis Roads)

Reduced Listening

Schaeffer's *reduced listening* theory is more concerned with the quality of sounds than their source.

What is striking about the Schaefferian theory, and what differentiates it from most music theory that developed before it, is its focus on the *primacy of the ear*, rather than abstract forms of musical construction.

Acousmatic listening is the opposite of direct listening, which is the 'natural' situation where sound sources are present and visible. By isolating the sound from the 'audiovisual complex' to which it initially belonged, it creates favourable conditions for reduced listening which concentrates on the sound *for its own sake*, as sound object.

Sound Object

The sound object "refers to every sound phenomenon and event perceived as a whole, a coherent entity, and heard by means of reduced listening which targets it for itself, independently of its origin or its meaning.... It is a sound unit perceived in its material, its particular texture, its own qualities and perceptual dimensions." It can be "compared to a `gestalt' in the psychology of form." (***Leigh Landy***)

"amongst several (sound) OBJECTS the PERMANENCE of a CHARACTERISTIC is the CONCRETE SONOROUS basis of a STRUCTURE Of VARIATIONS Of VALUE forming the ABSTRACT MUSICAL discourse." (***Michel Chion***)

Musique concrete, techniques

Sound transposition : reading a sound at a different speed.

Sound looping : repeat certain portions of a recording.

Reversal : playing the tape backwards

Comb Filtering : playing a sound against a delayed version of itself.

Splicing : change the order of the material, or insert new sounds.

Filtering : to bring in or out different frequencies of a recording.

Layering : multiple sources combined or by mixing them in real time.

Reverberation, delay : to create unity, fusion or space.

Categories of operation (Teruggi)

(1) Sound isolation and observation

Recording, listening to sound images.

(2) Sound editing

Cutting-out, incrustations, loops, time inversions, substitutions.

(3) Dynamic modifications

Amplitude modulation, actions on potentiometers, noise reduction or elimination, compression, expansion.

(4) Speed modifications

Speed variations, phase variations, Doppler effect.

(5) Time modifications

Time stretching, time contraction, time freezing, loops, time inversion.

Categories of operation (Teruggi)

(6) Spectral modifications

Filtering, resonant filtering, harmonisation, ring modulation, spectral interpolation, analysis-resynthesis.

(7) Density modifications

Shuffling, feedback, multiplication.

(8) Order of events modifications

Shuffling, editing, sound inversions.

(9) Space modifications

Panning, circling, Doppler effect, reverberation.

(10) Sound addition

Mixing, sound interleaving.

Parmegiani - Etude Elastique (1975)



Playing back

Sounds are usually loaded to buffers and played from there. Another approach is to stream sounds directly from disk. The first approach offers more flexibility but the second one consumes less memory resources.

Important parameters when playing back sounds are:

- * ***Speed*** of playback
- * ***Direction*** the samples is read in
- * ***Location*** where the reading takes place

SuperCollider allows for various ways to read buffer contents at different speeds, directions and locations.

Extending

Various methods exist to extend sounds, some examples:

- * ***Back to back*** reverse splicing of a sound or a succeeding one (Fugu fish).
- * ***Repetitions*** of sound file or a part of sound file to create controlled looping or rhythmic sequencing.
- * ***Layering sounds*** and creating "chords" by varying the speed of layered copies.
- * ***Overlapping*** a sound with itself where the copies have different start and end times.

Looping

Looping is a common procedure to create continuation. It can be used for example to pulsate a sound in a regular way

Very ***small loops*** can produce waveforms and continuous pitches.

Longer loops can be used for rhythmic purposes or meso level developments.

Looping parts of materials (and possibly to crossfade between loop instances) can help to prolong certain sounds.

Layered loops with slightly deviated loop times can cause interesting synchronization effects.

Brassage

A procedure which **chops** a sound into a number of **segments** and then redistributing these together according to a defined rule.

Brassage may be used for changing the **direction** of a sound, for **evolving montage** based on a source-sound, and for creating sample-based **textures**.

The **rule based** nature of the method provides an interesting path for algorithms to be used in composition.

Common application include **extending** the sound, **shuffling** its contents, texturize by **overlapping segments** and using procedures such as **random walks** for traversing a recorded sound.

Micromontage

Micromontage is the use of montage or detailed sequencing on the time scale of ***microsounds***.

Horacio Vaggione has used the method extensively in works such as ***Thema*** (1985), and ***Schall*** (1995).

The technique often makes use of extraction and arrangement of sound particles from a sample but also the creation and exact placement of sonic particle to create complex sound patterns or textures.

Micromontage may be accomplished through ***graphic editing***, a ***script***, or ***automated*** through a computer program.

Sequence and Texture

Trevor Wishart has identified two main categories of sound groups, sequences and textures.

A **sequence** has two general properties, a **field** and an **order**. The field defines the possible elements of a sequence and the order determines how these elements appear in time.

A **texture** differs from continuous sounds in that it consists of many discrete events.

A texture also has a **field** defining the possible events happening but it does not have a **clear order** of events and is rather defined by **density** and **distribution** properties.

Sound Transformation

Trevor Wishart has developed many methods of sound-transformation. General operations on recorded sounds include:

- * ***Synchronizing attacks*** in chords of the same sound.
- * ***Sound shrinking***, where a sound is repeated and then shortened on each repetition.
- * ***Zigzagging***, where sound is read backward-forward repetition and each iteration starts where the previous ended.
- * ***Doublets***, where where a sound is divided into repeating segments which are repeated and later respliced together.
- * ***Ceracu***, where segments repeat in several cycles that synchronize later after specified counts.

Curtis Roads

Curtis Roads has identified additional treatment of sounds

- * ***Balance mixing.*** Combine tracks while retaining the identity and individuality of each one.
- * ***Fusion mixing.*** Blend a low- level sound in order to fuse it with another sound.
- * ***Envelope reshaping.*** Scale a sound according to a drawn envelope function.
- * ***Tremolo.*** Slow amplitude modulation.
- * ***Pitch bending.*** Continuous pitch alteration in time by a variable factor.

Sound Transformation

One use of sound transformation can be as a compositional methodology, a certain form of **musical progression** where new material is derived from old. Here, **transformations** serve the structural function of variation.

"Just as states of matter undergo critical phase transitions from solid to liquid to gas to plasma, sound phenomena can pass from one state to another. These zones of morphosis are of extreme interest from an aesthetic point of view, as they are intrinsically exciting and fascinating. Zones of morphosis can be seen as the places where a continuous change in a plastic medium confronts context dependent thresholds of human perception. They occur when a change in quantity in some parameter appears as a qualitative change to the listener."

(Curtis Roads, Composing Electronic Music)

Effects

Processing

Alteration of sound signals through designed processes.

The following will be discussed:

- * *Delay*
- * *Reverb*
- * *Resonance*
- * *Distortion*
- * *Waveshaping*
- * *Dynamics*
- * *Phase vocoding*
- * *Resonators*

Delay

Delay effects can be categorized to **fixed** and **variable** delay effects. For fixed the delay time does not change. Three different thresholds can be defined for fixed delay times:

- * **Short** (less than 10 ms) mostly used for frequency-domain purposes for example to create filter.
- * **Medium** (10 to 50 ms) can give power to a 'thin' signal and create resonances.
- * **Long** (greater than 50 ms) produce echo outputs, repetitions and bouncing effects.

Delay

Variable time delay effects take place when the delay time is constantly changing.

Examples for variable delays are from effects of the 1960s and 1970s:

- * **Flanging** is when noise is heard in a mixture of direct and delayed sound. It is created by varying the delay time between 10 to 30 ms with a sine wave 0.1 to 20hz.
- * **Phasing** is similar to flanging but introduces filters to the signal chain and therefore a slightly less harsh sound.
- * **Chorus** consists of bank of variable delays with a slightly longer delay time than the ones mentioned above.

Reverb

Used to add *impression of space* to a sound source.

Reverb can be divided to three phases:

- * *Direct sound*
- * *Early reflections*
- * *Gradual reverberation*

The most common reverb technique in digital acoustics is to use *parallel comb delays* that are fed into few *allpass delays*.

Reverb

Popular parameters for controlling reverb algorithms are often:

- * Reverb *decay time*.
- * *Size* of emulated space.
- * High- and Low-frequency *damping* parameters.
- * *Pre-delay* time.
- * Amount of *early reflections*.

Convolution

Convolution is used both for *filters* and *impulse response based reverbs*.

The convolution algorithm **multiplies** an incoming source with an impulse response and transfers the characteristics of the impulse response onto the source sound.

Two main methods exist to record impulse responses, by recording an **impulse-like sound** (clapping or exploding a balloon) in the corresponding space or **playing back a sine-sweep** covering the whole frequency spectrum.

Other applications include **transferring the spectral envelope** (or other attributes) from one sound to the other with **cross-synthesis**.

Distortion

Distortion (or warping) is the alteration of an original signal by reducing or shaping it to produce harshness or overtones.

Examples include:

- * ***Saturation***, where levels are pushed above the maximum possible range of a waveform which introduces clipping in the resulting sound.
- * ***Bit reduction***, where the precision of a signal is lowered to such an extent that noises appear.
- * ***Feedback distortion***, where sound is fed back into the signal chain until distortion effects and sidebands result.

Waveshaping

Waveshaping is a type of distortion technique where complex spectra are produced altering the shape of the waveforms with other *(shaping or transfer) functions*.

The output of a waveshaping procedure is often a more **complex** sound than before and by modulating the **waveshaping index** (input amplitude) one can control the generated richness.

The transfer function is often **nonlinear** and the output hard to predict.

Compressor

A compressor **reduces** the **dynamic range** of a signal when it exceeds a certain **threshold** of amplitude.

The compression **ratio** determines how much the signal that exceeds the threshold is turned down. 2:1 compression ratio means that for every 2 dB of signal that goes into the unit, only 1 dB comes out.

A **sustainer** works like an inverted compressor, it exaggerates the low amplitudes and tries to raise them up to the threshold defined.

Limiters

The **limiter** operates in a similar way as a compressor does, but looks at the signal's peaks whereas the compressor looks at the average energy level.

A limiter will **not let the signal past** the threshold, while the compressor does, using the compression ratio.

Limiters can be considered as more extreme cases of compression with the main difference being the **lower ratios** used in compression and the operation of a **maximum setting** for a limiter.

Noise Gate

The **noise gate** allows a signal to pass through the filter only when it is above a certain threshold.

When the signal level is **below the threshold**, no sound is allowed to pass.

A noise gate usually has **attack** and **release** settings to control how the gate kicks in.

Noise gate is often used to **remove background noise** but can also create interesting **rhythmic effects** as well as to **cross-modulate** signals.

Phase Vocoding

Phase vocoding is a form of transformation via analysis where a sound file is analyzed to analysis bands and is then resynthesized by the phase vocoder.

The algorithm allows **frequency-domain modifications** to the resulting analysis data (typically time expansion/compression and pitch shifting).

Many possible operations such as:

- * **Shuffling** of analysis data
- * **Morphing** two sets of analysis data
- * Make **spectral gaps** in analysis data
- * **Freeze** spectral segments
- * **Smooth** out analysis data

Resonator

A resonator ***adds pitch or oscillation*** to an input signal for specified resonant frequencies.

Resonators are usually created ***using short delay lines*** with ***feedback*** and often occur in ***banks*** where several ones are used simultaneously.

Can be used to ***give pitch*** to input sounds, make ***chords*** from input sounds or ***tune sounds*** to the frequencies of others with subtle resonating behaviors.

```

(
  NF(\iop, {|freq=78, mul=1.0, add=0.0|
    var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
    var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
    var out = DFm1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
    HPF.ar(out, 40)
  }).play;
)

(
  NF(\dsc, {|freq = 1080|
    HPF.ar(
      BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
        SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
        LFNoise1.ar([12,14,10]).range(100,900),
        SinOsc.ar(20).range(9,11)
      ), 80)
    ).play;
)

var <>pindex, <>cindex;

initialize {
  if(pindex.isNil, { pindex = 1000 });
  if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
  pindex = 1000;
  (this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
  if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
  if(index.isNil && pindex.isNil, {
    this.initialize();
  });

  pindex = pindex + 1;
  this[pindex] = \filter -> process;
}

control {|process, index|
  var i = index;

  if(i.isNil, {
    this.initialize();
    cindex = cindex + 1;
    i = cindex;
  });

  this[i] = \pset -> process;
}

(
  NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
    var trig, seq, freq;
    trig = Dust.kr(rate);
    seq = Diwhite(freqMin, freqMax, inf).midicps;
    freq = Demand.kr(trig, 0, seq);
    HPF.ar(PMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
      LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
  }).play;
)

```

Exercises

Exercises

1. Create a sound that plays at the same time in reverse and in a normal direction. The reverse should be pitch shifted down while the normal up.
2. Create a sound where two loops are heard but with different lengths.
3. Implement a sample playing process where the rate of playback goes from one octave down to an octave up over 10 seconds.
4. Create a sound process that oscillates between a pitched down sound sample and pitched up one.
5. Implement a pattern that will play chords using sound samples as base material.
6. Implement a pattern that will play bits of a sound file somehow randomly while starting at the beginning and progressing towards the end of the file.

Exercises

1. Create a reverb based on a network of all pass filters where each component is slightly distorted.
2. Create an echo effect that uses different processing for low and high frequencies.
3. Create an effect based on delays where the delay time is changing.
4. Experiment with a network that includes a resonator, filters and distortion to emphasise important frequencies in an input sound.

